



Apex Institute of Technology

Computer Science & Engineering

Experiment 7

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Subject Name: Database
Management System

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AIM: To design and implement a **materialized view** and to **compare and analyze execution time** and performance differences between **simple views**, **complex views**, and **materialized views**, thereby understanding their impact on query optimization and system performance.

OBJECTIVES:

- To create **simple views**, **complex views**, and **materialized views**, and to evaluate their performance by comparing query execution time for each, highlighting the advantages of materialized views in enterprise-level applications.

SOFTWARE REQUIREMENTS:

- Database Management System:
 - PostgreSQL Database
- Database Administration Tool / Client Tool:
 - pgAdmin

PRACTICAL/EXPERIMENT STEPS:

1. Departments and employees tables were created in the database to store department details and employee information.
2. Sample records were inserted into both tables to provide data for testing views.
3. A simple view was created on the employees table using a salary filter to display high-paid employees.
4. A complex view was created using JOIN and aggregation functions to calculate department-wise employee count and average salary.
5. A materialized view was created to store the precomputed results of the complex query.



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6. SELECT queries were executed on all three views to observe the output.
7. EXPLAIN ANALYZE was used to compare the execution time and performance of simple, complex, and materialized views.

PROCEDURE:

1. The pgAdmin (PostgreSQL) environment was opened to access the database.
2. Departments and employees tables were created with appropriate fields and relationships using PRIMARY KEY and FOREIGN KEY constraints.

```
CREATE TABLE  
  
Query returned successfully in 444 msec.
```

3. Sample data was inserted into both tables using INSERT statements.

```
INSERT 0 3  
  
Query returned successfully in 153 msec.
```

4. A simple view was created to display employee names and salaries above a specified threshold.
5. A complex view was created using JOIN and GROUP BY to calculate department-wise statistics.

	dept_name character varying (20) 🔒	total_employees bigint 🔒	avg_salary numeric 🔒
1	Finance	1	55000.000000000000
2	IT	3	86000.000000000000
3	HR	2	62500.000000000000

6. A materialized view was created to store the results of the complex query for faster access.

	dept_name character varying (20) 🔒	total_employees bigint 🔒	avg_salary numeric 🔒
1	Finance	1	55000.000000000000
2	IT	3	86000.000000000000
3	HR	2	62500.000000000000

7. Queries and EXPLAIN ANALYZE statements were executed to observe results and compare performance.



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CODE:

```
CREATE TABLE departments(  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(20)  
)
```

```
CREATE TABLE employees(  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(20),  
    dept_id INT REFERENCES departments(dept_id),  
    salary NUMERIC  
)
```

```
INSERT INTO departments VALUES  
(1, 'IT'),  
(2, 'HR'),  
(3, 'Finance')
```

```
INSERT INTO employees VALUES  
(101, 'Amit', 1, 90000),  
(102, 'Riya', 1, 80000),  
(103, 'Karan', 2, 60000),  
(104, 'Neha', 2, 65000),  
(105, 'Arjun', 3, 55000),  
(106, 'Priya', 1, 88000)
```

```
-- simple View  
CREATE VIEW simple_view AS  
SELECT emp_name, salary  
FROM employees  
WHERE salary > 70000
```

```
SELECT * FROM simple_view
```

```
-- complex view  
CREATE VIEW complex_view AS  
SELECT d.dept_name,  
    COUNT(e.emp_id) AS total_employees,  
    AVG(e.salary) AS avg_salary  
FROM employees e
```



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```
JOIN departments d  
ON e.dept_id = d.dept_id  
GROUP BY d.dept_name
```

```
SELECT * FROM complex_view
```

```
-- materialised view  
CREATE MATERIALIZED VIEW materialized_view AS  
SELECT d.dept_name,  
       COUNT(e.emp_id) AS total_employees,  
       AVG(e.salary) AS avg_salary  
FROM employees e  
JOIN departments d  
ON e.dept_id = d.dept_id  
GROUP BY d.dept_name
```

```
SELECT * FROM materialized_view
```

```
REFRESH MATERIALIZED VIEW materialized_view
```

```
EXPLAIN ANALYZE SELECT * FROM simple_view  
EXPLAIN ANALYZE SELECT * FROM complex_view  
EXPLAIN ANALYZE SELECT * FROM materialized_view
```

I/O ANALYSIS:

1. Simple View Output

Displays employee names and salaries where salary is greater than 70000.

	emp_name character varying (20)	salary numeric
1	Amit	90000
2	Riya	80000
3	Priya	88000

2. Complex View Output

Displays department-wise data including total employees and average salary.

	dept_name character varying (20)	total_employees bigint	avg_salary numeric
1	Finance	1	55000.000000000000
2	IT	3	86000.000000000000
3	HR	2	62500.000000000000



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3. Materialized View Output

Displays the same aggregated results as the complex view but retrieved faster.

	dept_name character varying (20)	total_employees bigint	avg_salary numeric
1	Finance	1	55000.000000000000
2	IT	3	86000.000000000000
3	HR	2	62500.000000000000

4. Performance output

EXPLAIN ANALYZE shows execution time, where the materialized view executes faster than the complex view.

	QUERY PLAN text
1	Seq Scan on employees (cost=0.00..18.00 rows=213 width=90) (actual time=0.054..0.058 rows=3.00 loop...
2	Filter: (salary > '70000'::numeric)
3	Rows Removed by Filter: 3
4	Buffers: shared hit=1
5	Planning Time: 0.313 ms
6	Execution Time: 0.088 ms

	QUERY PLAN text
1	HashAggregate (cost=53.14..55.64 rows=200 width=98) (actual time=0.137..0.144 rows=3.00 loops=1)
2	Group Key: d.dept_name
3	Batches: 1 Memory Usage: 32kB
4	Buffers: shared hit=2
5	-> Hash Join (cost=30.25..48.34 rows=640 width=94) (actual time=0.102..0.111 rows=6.00 loops=1)
6	Hash Cond: (e.dept_id = d.dept_id)
7	Buffers: shared hit=2
8	-> Seq Scan on employees e (cost=0.00..16.40 rows=640 width=40) (actual time=0.033..0.034 rows=6.00 loops=1)
9	Buffers: shared hit=1
10	-> Hash (cost=19.00..19.00 rows=900 width=62) (actual time=0.035..0.036 rows=3.00 loops=1)
11	Buckets: 1024 Batches: 1 Memory Usage: 9kB
12	Buffers: shared hit=1
13	-> Seq Scan on departments d (cost=0.00..19.00 rows=900 width=62) (actual time=0.023..0.025 rows=3.00 loo...
14	Buffers: shared hit=1
15	Planning Time: 0.336 ms
16	Execution Time: 0.240 ms



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	QUERY PLAN	
	text	
1	Seq Scan on materialized_view (cost=0.00..16.40 rows=640 width=98) (actual time=0.025..0.027 rows=3.00 loo...	
2	Buffers: shared hit=1	
3	Planning Time: 0.082 ms	
4	Execution Time: 0.044 ms	

LEARNING OUTCOMES:

1. Understood the concept and working of materialized views in a database system.
2. Differentiated between simple views, complex views, and materialized views.
3. Created materialized views in PostgreSQL.
4. Compared query execution time for different types of views.
5. Analyzed performance benefits of materialized views in data-intensive applications.
6. Applied materialized view concepts in real-world company scenarios.